BAHAA EBRAHEEM

Senior Full Stack & AI Engineer | .NET · Python · LLMs · AI Agents

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SUMMARY

Senior Full Stack & AI Engineer with 7+ years building and operating production systems on ASP.NET Core, React, Python and PostgreSQL / SQL Server, much of it delivered fully remote for teams in the UAE and Iraq. Ships production AI end to end: LLM features on Google Gemini, OpenAI and Anthropic Claude behind one provider-agnostic layer, tool-calling AI booking agents on WhatsApp and Telegram, AI image generation pipelines, and a bilingual Arabic/English NLP service (Python, FastAPI, Transformers) that screens CVs with an F1 of 0.986 against 0.707 for a keyword ATS. Also shipped a national e-government platform live at register.mot.gov.iq.

KEY ACHIEVEMENTS

- **Multi-LLM platform in production —** New Look runs Google Gemini, OpenAI and Anthropic Claude behind one provider-agnostic AI layer; providers and models switch from an admin registry with no redeploy, prompts are admin-editable templates, and slow image generations run in a background job queue so the UI never blocks.
- **Bilingual AI CV screening (MSc thesis) —** Chance4Hire's Python/FastAPI NLP service ranks candidates with a multilingual BERT sentence encoder; on a 40-CV gold set it reaches F1 0.986 vs 0.707 for a naive keyword ATS, and lifts Arabic skill recall from 26.5% to 100%. AI service is open source on GitHub.
- **AI booking agents on WhatsApp & Telegram —** integrated New Look with Callivox: tool-calling LLM bots on WhatsApp and Telegram that book, reschedule, cancel and delete appointments (including group bookings) and answer availability, service, invoice, payment and account questions through 16 secured API tools with client verification, plus the Callivox inbox embedded in the staff app.
- **Zero-discrepancy migration off Fresha —** built the tool that moved a live salon's full history from Fresha — clients, visits, service lines, invoices, payments, gift cards, packages and stock; an automated audit matched all 17 categories with zero differences.
- **Production storage migration —** moved 162,516 files (223 GB) off a nearly-full production disk onto S3-compatible object storage with a real outage of only 7–10 minutes. SHA-256 deduplication proved 38.4% of stored bytes were duplicates, cutting the upload to 138 GB. A rehearsal surfaced 9 defects, all fixed before go-live.
- **National e-government platform —** Iraq's Companies Registrar, live at register.mot.gov.iq: a no-code workflow engine whose visual designer lets registrars compose 40+ licensing services without writing code.
- **Storage microservice with no database —** takes bytes, returns one opaque storage key. MinIO and Ceph RADOS Gateway share one code path, so the backend changes by configuration with zero data migration.
- **Remittance platform in national production —** EMPS moves money for Haram Internal Transfers across every one of its service points in Syria. Built MISAgent, the Windows desktop bridge deployed at those points, giving the Blazor WebAssembly client hardware the browser sandbox cannot reach — barcode scanners, label printers and flatbed scanners — streamed into the remittance workflow in real time.
- **Security —** found and closed a SignalR broadcast that leaked every user's events to every connected client.

TECHNICAL SKILLS

- **Backend:** C#, ASP.NET Core (7–10), ASP.NET MVC, Microservices, SignalR (WebSockets), REST API, GraphQL, ABP Framework, Entity Framework Core, LINQ, Node.js / Express.
- **Frontend:** React 19, Vite, Tailwind CSS, TanStack Query, Blazor (WASM/Server), Angular, TypeScript, JavaScript, RxJS, Syncfusion, Bootstrap, DevExpress.

- **AI & LLM:** LLM Integration (Google Gemini, OpenAI GPT, Anthropic Claude), AI Agents & Tool Calling, Prompt Engineering, Multi-provider AI Architecture, AI Image Generation, Background AI Jobs.
- **NLP & Machine Learning:** Python, FastAPI, Transformers / BERT, Sentence-Transformers, Hugging Face, PyTorch, Embeddings & Semantic Similarity, Arabic NLP, Document AI (PDF text extraction), Model Evaluation (Precision / Recall / F1, ablation).
- **Integrations:** Callivox, WhatsApp Business API, Firebase Cloud Messaging, Webhooks, OpenTelemetry.
- **Mobile & Desktop:** Flutter (Android & iOS), Firebase push notifications, Electron (Windows desktop), Windows Forms (C#).
- **Databases:** SQL Server, PostgreSQL, Redis, MongoDB, Oracle, ADO.NET.
- **DevOps & Cloud:** Docker, Azure DevOps, AWS, GitHub, CI/CD Pipelines, TFS, PowerShell, MinIO, Ceph RADOS Gateway, S3-compatible Object Storage, Prometheus, Grafana, Serilog.
- **Architecture & Domain:** Clean Architecture, CQRS, Domain-Driven Design, Double-entry Accounting (Business Central model), Data Migration, Unit Testing (xUnit), Agile/Scrum.

PROFESSIONAL EXPERIENCE

Full Stack Developer (AI & LLM focus) | HA Consultancy 06/2026 – Present | UAE (Remote)

- New Look — AI-powered salon-management platform (a Fresha alternative) on ASP.NET Core 10, React 19 / TypeScript, PostgreSQL and SignalR: multi-branch role-based access, WhatsApp Business API, and an Electron chair-station with an OBSBot camera.
- **AI integration (Gemini, OpenAI, Claude) —** Gemini Pro + image model generate 8 personalized hairstyle suggestions from 6-angle client photos, with AI face-shape analysis, consultation chat, service plans and package recommendations; one provider-agnostic layer with a background AI job queue and admin-editable prompts.
- **Callivox AI agents (WhatsApp & Telegram) —** tool-calling LLM bots that book, reschedule, cancel and delete appointments and answer availability, invoice and account questions, over 16 secured public API tools with an API-key gate and client verification by code; Callivox inbox embedded in the app.
- **Fresha migration —** migration tool with an automated audit (C# + Python) that moved the live salon's full history off Fresha with zero differences across 17 audited categories.
- **Database audit trail —** 94 deferred PostgreSQL triggers capture every change with the user, route and request id into an outbox; a background shipper moves it, idempotently and with SKIP LOCKED, into a separate month-partitioned audit database.
- **Commission engine —** fixed and tiered commission ladders; a set-based SQL backfill rebuilt the full commission history on production in 2 minutes instead of 4.6 hours, after a real dry run inside a rolled-back transaction.
- **Accounting system (Business Central model) —** general ledger with chart of accounts, account categories, parent/child hierarchy and opening balances; a journal engine with templates, approval, posting and reversal that keeps the audit history; and configurable posting groups that map items, service categories, customer groups and payment methods to G/L accounts, with posting validation and restrictions.
- **Mobile app (Android & iOS) —** Flutter app for barbers and clients: role-based sign-in, appointment calendar, and Firebase push notifications with real-time updates over SignalR.
- Online Booking Portal — public self-service booking: clients pick services, stylist and time slot, synced in real time into the main schedule; React 19 / TypeScript, RTL Arabic/English.

Senior Full Stack Engineer | Almaseer Tech 06/2026 – 09/2026 | Baghdad, Iraq (Remote, Contract)

- Iraqi Companies Registrar — Ministry of Trade — National e-government platform on a no-code workflow engine: a visual React Flow designer lets registrars compose 40+ multi-step licensing services without writing code. ASP.NET Core 8, React 18 / TypeScript, SQL Server and Redis, fully RTL Arabic UI. Reworked the payment model around per-node fee selection.

Senior Full Stack Engineer | NETAQ For E-Solutions 10/2025 – 06/2026 | Abu Dhabi, UAE (Remote)

- **Emirates Arabian Horses Society** — Scalable backend on ASP.NET Core 10 with Umbraco 17 CMS, a React frontend, and Flutter mobile apps.
- **Stallions Semen Market** — High-performance ASP.NET Core MVC backend using idempotency keys, row-level locking and concurrency-safe transactions to guarantee accurate dose selling under high traffic.
- **Abu Dhabi Excellence Award Portal** — Scalable backend on ASP.NET Core 10 with Umbraco 17 CMS as a toggleable per-client plugin, a React frontend, a GraphQL unified API layer, and OpenTelemetry tracing.
- **Award and recognition platforms (ASP.NET Core MVC) —** built and enhanced submission, evaluation and workflow modules for Khalifa Award For Education, Fatima Bint Mubarak Women Sport Award, Individual Excellence Medals Award, and Kanz Al Jeel.

Senior Full Stack Developer | MIS Solutions 09/2022 – 10/2025 | Damascus, Syria

- **EMPS — Remittance Management System —** Microservices-based remittance platform on ASP.NET Core 7 and ABP, live at Haram Internal Transfers, with Blazor WebAssembly and SignalR delivering instantaneous transaction updates.

Backend Developer | Solutions LLC 01/2019 – 06/2022 | Dubai, UAE (Remote)

- **Built and optimized enterprise ASP.NET MVC applications** and SQL Server schemas, significantly improving data-retrieval performance.

FEATURED PROJECTS

- **Chance4Hire — AI recruitment platform & MSc thesis: bilingual (Arabic/English) CV screening and candidate ranking.** AI service (public): github.com/BahaaEbraheem/chance4hire-ai-service
  - **NLP / Transformer matching —** Python + FastAPI microservice; a multilingual BERT sentence encoder (paraphrase-multilingual-MiniLM-L12-v2 on Sentence-Transformers / PyTorch) scores CV-to-job fit semantically, cross-lingual and paraphrase-tolerant, while skill extraction stays explainable through a bilingual gazetteer with Arabic normalization.
  - **Measured, not claimed —** evaluation harness on a 40-CV gold set (142 skills; Arabic, English and code-switched): F1 0.986 vs 0.707 for a naive keyword ATS and 0.767 for a tuned one, Arabic recall 100% vs 26.5%, an ablation study, error analysis and latency tracking (avg 530 ms).
  - **Document AI —** Arabic-aware PDF extraction (PyMuPDF recovers 12/13 skills vs 8/13 with pdfplumber), scanned-PDF detection, education and years-of-experience extraction including Arabic number words, and profile auto-fill from the uploaded CV.
  - **AI in production —** the .NET backend calls the AI service through a resilient analyzer that falls back to a local engine if the service fails; live analysis progress streams over SignalR; recruiters keep the final decision with notes and feedback, and an accuracy dashboard reports the model's metrics.
  - **Platform —** ASP.NET Core / ABP (DDD), React 19, PostgreSQL, Redis, Docker; multi-company isolation proven by tests; search latency cut from 117 ms to 24 ms with pg_trgm GIN indexes.
- **Webhook Service:** A scalable notification engine built with **ASP.NET Core 9** and **Angular**. Features advanced caching, indexing, and is fully **Dockerized**.
- **Library Management System:** A full-stack solution featuring a **Flutter** mobile app and a **Node.js /Express** backend with **MongoDB** and **JWT** security.

EDUCATION

- **MSc Web Science** — Syrian Virtual University, 2024 · BSc Software Engineering — Aleppo University, 2016
- **Languages:** English — Proficient (Technical & Professional) · Arabic — Native